

HAT ISLAND, NU, Canada

WMO: 710840

Lat: 68.320N

Lon: 100.090W

Elev: 118

StdP: 14.63

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.8	-38.6	N/A	N/A	N/A	N/A	N/A	N/A	42.5	-18.9	37.5	-16.2	12.9	270	1.230

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	58.2	N/A	53.8	N/A	50.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.9	33.6	29.6	DB	-43.5	65.6	3.8	4.6	-46.2	68.9	-48.5	71.6	-50.6	74.2	-53.4	77.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-23.1	-24.8	-18.1	-3.8	16.4	34.2	42.5	42.0	33.8	19.9	-1.2	-16.0	(6)	
	DBStd	26.40	9.67	9.69	11.33	10.46	9.39	6.23	5.64	5.64	5.91	8.30	10.72	10.99	(7)	
	HDD50	15107	2267	2094	2112	1613	1040	477	245	260	486	933	1535	2045	(8)	
	HDD65	20557	2732	2514	2577	2063	1505	925	697	715	936	1398	1985	2510	(9)	
	CDD50	25	0	0	0	0	0	2	13	10	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	16.0	16.3	15.9	15.8	15.8	15.4	14.3	13.9	14.6	17.3	18.4	17.5	16.6	(14)	
	PrecAvg	6.7	0.2	0.2	0.3	0.3	0.3	0.6	1.0	1.1	1.1	0.7	0.4	0.4	(15)	
(16) Precipitation	PrecMax	9.3	0.8	0.5	1.6	0.7	0.9	1.5	2.1	2.2	2.2	1.7	1.0	0.8	(16)	
	PrecMin	4.2	0.0	0.0	0.1	0.1	0.0	0.1	0.4	0.5	0.6	0.2	0.1	0.0	(17)	
	PrecStd	1.4	0.2	0.1	0.3	0.1	0.2	0.3	0.4	0.4	0.4	0.4	0.2	0.2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	7.0	15.7	28.1	35.9	58.7	66.2	63.5	51.0	35.8	27.8	20.6	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	2.8	1.6	9.5	21.7	33.3	51.2	59.7	57.3	46.8	33.5	21.9	11.9	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	-3.4	-2.9	5.3	17.2	32.0	46.0	54.8	53.2	44.1	31.9	18.3	6.3	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	-8.8	-8.7	0.0	12.4	29.7	42.0	51.2	49.7	41.9	30.4	14.2	0.5	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.2	11.8	13.3	10.2	7.5	10.8	8.0	5.3	6.1	9.6	10.0	(35)	
		MCDBR	15.5	16.4	15.6	15.3	9.8	14.5	17.5	13.9	8.5	5.6	11.8	15.9	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.150	0.207	0.238	0.292	0.313	0.327	0.357	0.329	0.301	0.245	0.156	N/A	(40)	
	taud		1.873	2.042	2.175	2.051	2.045	2.166	2.263	2.509	2.514	2.207	1.891	N/A	(41)	
	Ebn at Noon		40	192	258	268	275	275	261	260	239	172	31	0	(42)	
	Edh at Noon		2	15	24	37	43	39	35	24	19	12	2	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		9	156	663	1418	2070	2127	1794	1136	534	171	28	0	(44)	
	RadStd		1	9	30	57	102	123	113	86	67	28	3	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page